Evaluating Computational Principles in the Brain: A Replication Study and Analysis of Words Effects on Neural Encoding

Candidate Number: 6243J

Course: AI and Neuroscience

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Word Count: 1498 words

Introduction

The influential paper entitled *"Shared Computational Principles for Language Processing in Humans and Deep Language Models"* [1], provides empirical evidence that the human brain processes language in a similar way to auto-regressive deep language models. This project aims to replicate the study's key findings (figures 4 - 6) and presents a new hypothesis using the same dataset.

Methods

The original paper selected electrodes through a randomization procedure which is computationally demanding therefore, an alternative approach was used. Following the paper, electrodes only from the left-hemisphere were selected. A one-sample t-test was applied to the GloVe embedding correlations (averaged 10 folds cross-validation) to asses whether correlation were significantly greater than zero. False discovery rate (FDR) correction was applied ($q < 0.01$) resulting in 174 significant electrodes (See figure 1).

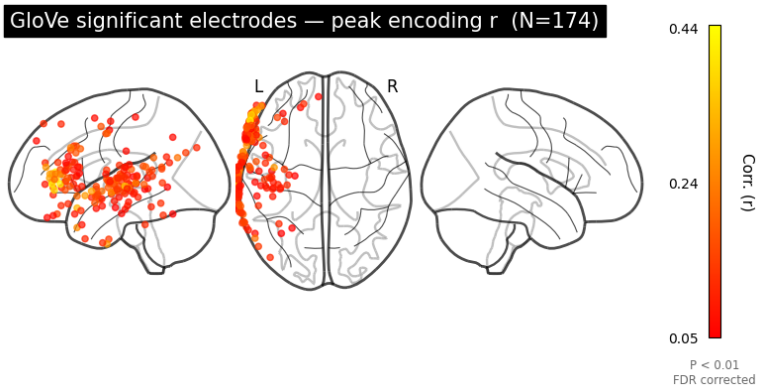
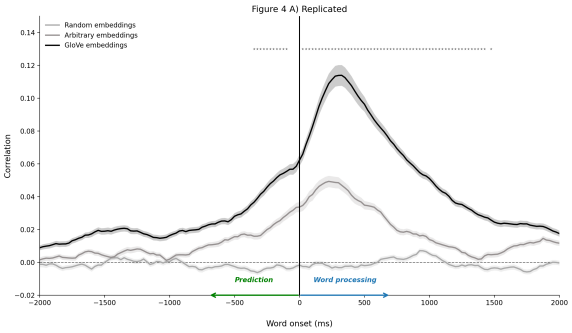


Figure 1: Plot of the all the electrodes selected across 9 patients and their GloVe encoding correlations.

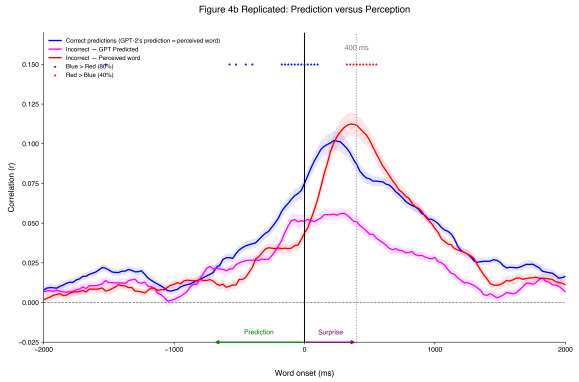
All encoding models used ridge linear regression. Following the paper and to reduce noise, signals were temporally smoothed using a 200ms moving average window with 25 ms intervals.

Results

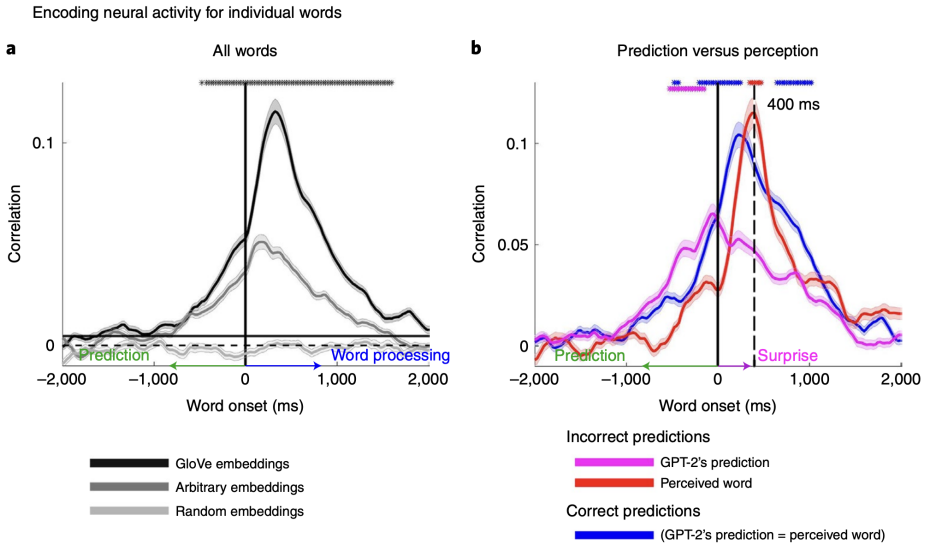
Figure 2 faithfully replicates the paper's Figure 4. GloVe embedding has the highest correlation compared to arbitrary and random embeddings (4a). Furthermore, the words that were incorrectly predicted decay after word onset (4b) whereas, the incorrect but perceived words peak at approximately 400ms post onset. This represents the post-onset surprise signal as correctly predicted words have lower activity and reach their peaks earlier.



(a) Replication: 4a



(b) Replication: 4b

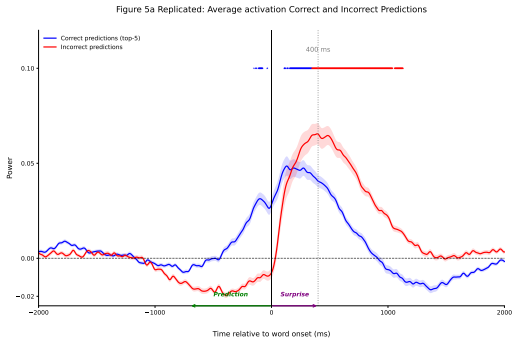


(c) Original figure 4 as reported by [1]

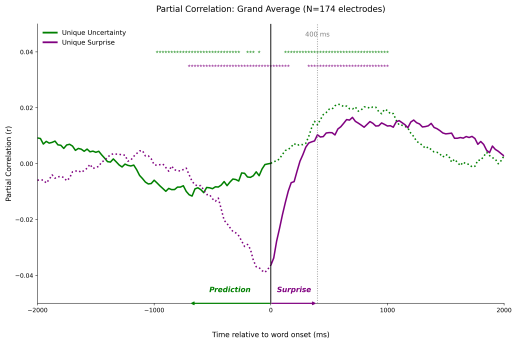
Figure 2: Comparison of figure 4 from the paper. Note for the replications, the asterisks indicate time points of significant differences in encoding correlation, identified via permutation testing with FDR correction and majority-electrode agreement ($> 50\%$).

Nevertheless, the replication presents a lower correlation for incorrectly predicted words pre-onset. This discrepancy likely reflects nosier electrode selection and different word filtering; 4,867 words compared to paper's original 4,843 despite using the same 50 dimensional GloVe embedding.

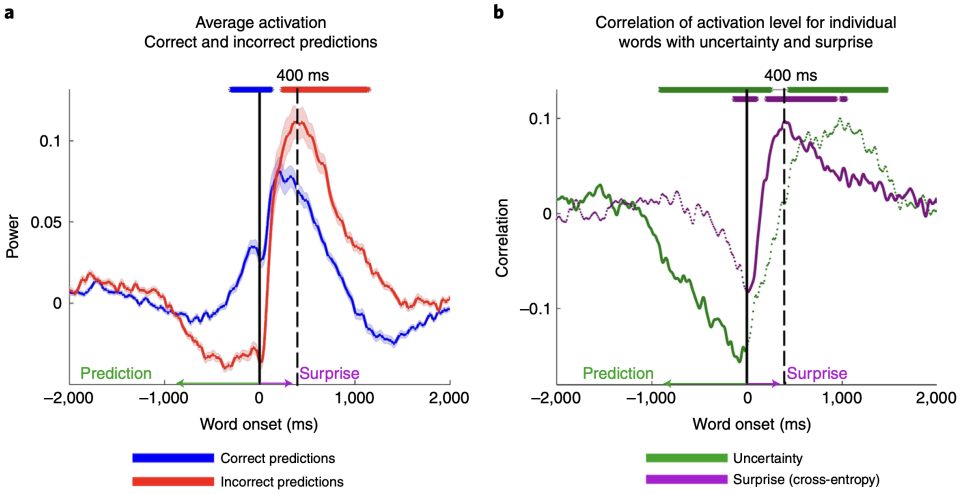
The morphology of figure 5a was successfully replicated in figure 3 as incorrect words show higher and slightly delayed peak activation compared to correct words. However, the power magnitude differs likely due to different pre-processing implementations. Figure 3 was plotted by first performing z-scoring and then applying hamming smoothing. Another possible explanation for the difference is electrode selection, as a different subset of electrode can influence magnitude.



(a) Replication: 5a



(b) Replication: 5b



(c) Original figure 5 as reported by [1]

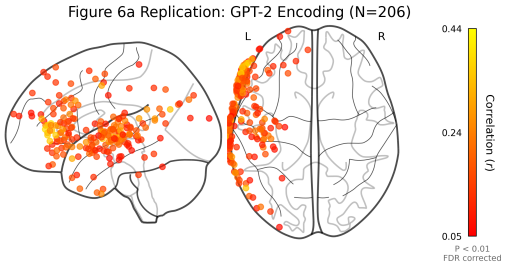
Figure 3: Comparison of figure 5 from the paper. Note for the replications, the asterisks indicate time points of significant differences in encoding correlation (same as figure 2).

These differences can also account for the discrepancies in the replication of figure 5b. In particular, the uncertainty signal does not show a negative correlation pre-onset, rather, it shows no correlation. Reducing the analysis to five electrodes shifted the power magnitude towards the paper's original level (see appendix figure 9), highlighting the influence of electrode selection. Differences in both the pre-processing of brain data and words-level features (tokenization-to-word alignment) can alter correlation results. Moreover, Pearson correlation shows negative pre-onset correlation (see appendix figure 8) indicating controlling for surprise reduces uncertainty's effects. Conversely, the surprise signal largely replicates the paper's pattern.

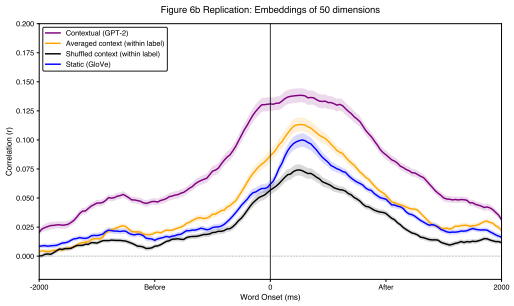
For replication of figure 6a, randomization procedure was not used due to computational constraints. Instead, electrodes were selected using the same procedure as before; one-sample

t-test with FDR correction on the average correlation using GPT embeddings. This yielded 206 significant electrodes (See figure 4). Despite fewer significant electrodes than in the paper, there was substantial overlap with the paper's brain regions and peak correlations were higher for GPT than for Glove embeddings.

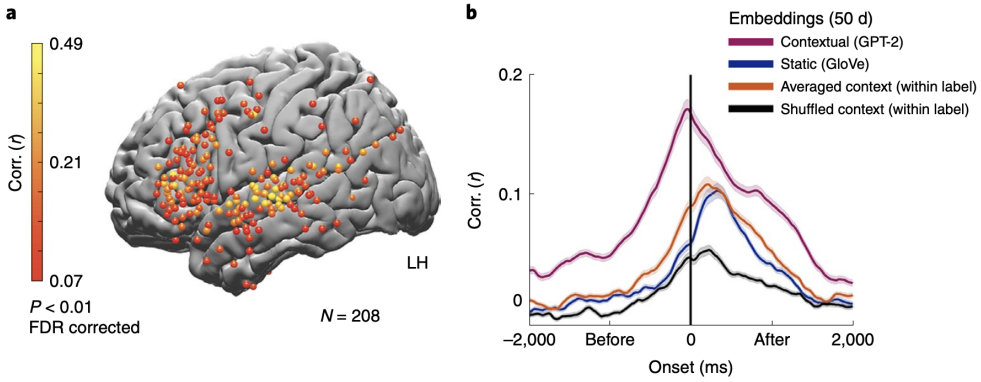
Similarly, GPT embeddings had the highest correlation in figure 6b replication, followed by average, GloVe, and Shuffled embeddings. These results align with the paper's figure 6b, although the slope for the GPT correlation is more gradual and the shuffled baseline has a higher correlation that expected. These differences likely stem from electrode and word selection.



(a) Replication: 6a



(b) Replication: 6b



(c) Original figure 6 as reported by [1]

Figure 4: Comparison of figure 6 from the paper; top subplots are replications and button subplots are the paper's original.

Novel Research Question

Are the shared computational principles between the human brain and language models influenced by semantic distance and word type? To investigate this, the following hypotheses are tested:

(i) **Graded Error Potentials:** for incorrectly predicted words, the neural surprise signal scales with the semantic distance between the predicted and actual word.

(ii) **Word Category Alignment:** correlation between neural signal and language model is non-uniform across linguistic categories.

Motivation and Methods

Predictive coding theory states the brain is a continuous prediction machine [2, 3]. Indeed, the paper's result align with this theory. However, the paper treats all input words equally, whereas, predictive coding suggests different inputs can have different brain signals [4]. An input does not produce a binary response, but rather a graded neural signal modulated by the magnitude of the prediction error [5].

Inspired by this theory and prior work [6], the first hypothesis examines the idea of a graded potential by quantifying how incorrect the incorrect GPT prediction was. Degree of incorrectness was determined by calculating the cosine similarity for the GloVe embedding of the predicted versus perceived word. A tertile split based on the cosine distance was used due to the right-skewed distribution (See figure 5). Intuitively, a semantically similar word (e.g. "gorilla" instead of "monkey") should produce a smaller post-onset surprise signal than an unrelated word.

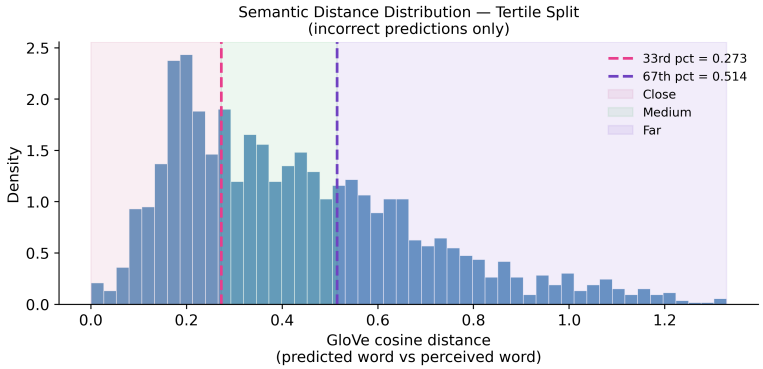


Figure 5: Histogram showing word density to Glove cosine distance. Tertile split was used due to right-skewed distribution (words tended to be closer to real prediction than further). The three splits were: less than of equal 0.27 (short distance; 660 words), more than 0.27 but less than or equal to 0.51 (medium distance; 659 words), and more than 0.51 (Far; 659 words).

The second hypothesis was also inspired by predictive codings suggestions that different inputs have different internal representations [7]. Instead of treating all words equally, words are grouped into 3 linguistic categories:

- Content words (words with meaning) versus Function words (e.g. the)
- Nouns versus Verbs
- Concrete (words perceived through senses e.g. apple) vs Abstract words (lack sensory form e.g. justice)

Several studies have found words from different lexical categories elicit different neural responses [8, 9, 10]. For example, one study found stronger activity in the dorsolateral frontal and lateral temporal cortex for verbs compared to nouns [8]. Although different word types are know to produce distinct neural responses, it is unclear whether this relationships is captured through an encoding analysis using GloVe and GPT embedding.

The podcast study [11] did evaluate syntactic embeddings but never grouped GloVe or GPT embeddings according to the word syntax. Nevertheless, the study's synaptic tags were used

to categorise words as verbs, nouns, functional, and non-functional. The Brysbaert concreteness dataset was used [12] to group words as either concrete (rating ≥ 4.0) or abstract (rating (≤ 2.5)) and functional words were omitted to ensure data balance.

Results and Discussion

Graded Error Potentials

The predictive signals prior to word onset (figure 6 subplot left) demonstrates a slight modulation by semantic distance as the smallest error group closely aligns with the correct predictions in contrast to the larger error groups. This reflects a greater neural encoding of predicted words that are semantically similar to the true word however, current evidence is insufficient to determine underlying mechanisms and warrants further study. What can be clearly de-

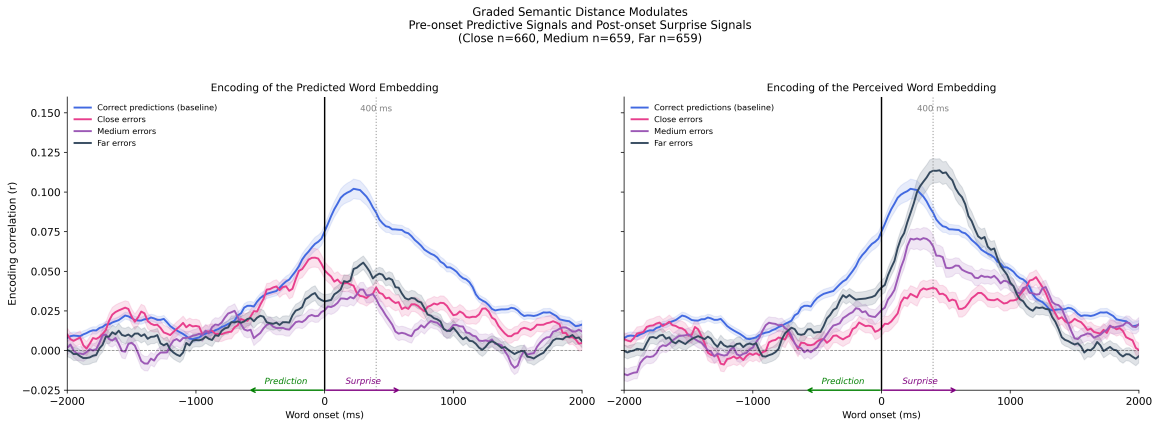


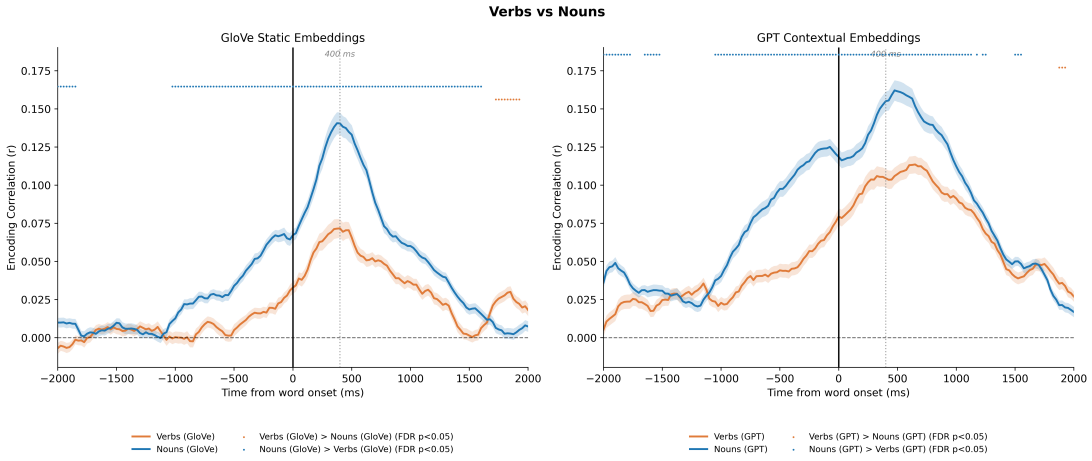
Figure 6: Encoding results for hypothesis 1: grouping words according to GloVe cosine similarity distance.

termined is a graded pattern post word onset for perceived word encodings (figure 6 subplot right). Words that were further from the prediction produced the largest neural response with a peak at 400ms which is consistent with semantic surprisal literature [13]. Medium and close error words showed progressively smaller responses with earlier peaks (see appendix figure 10), suggesting a greater neural prediction error and longer processing time for words that violate more strongly expectations.

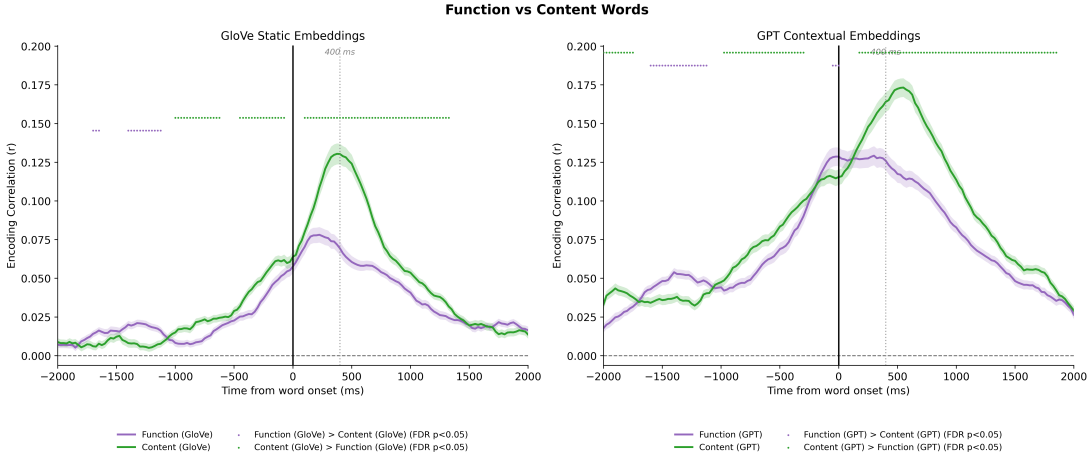
In contrast, predicted post onset words did not exhibit a clear graded pattern as the signals decayed similarly across error groups (see figure 6 subplot left). However, the smallest error group peaked earlier than the larger error groups, reinforcing the idea that words that are more similar to the brain's expectations are resolved more quickly than mismatching words. Overall, the results are consistent with the hypothesis made; the brain is sensitive to the degree of semantic mismatch rather than representing information in a binary correct-incorrect manner.

Word Category Alignment

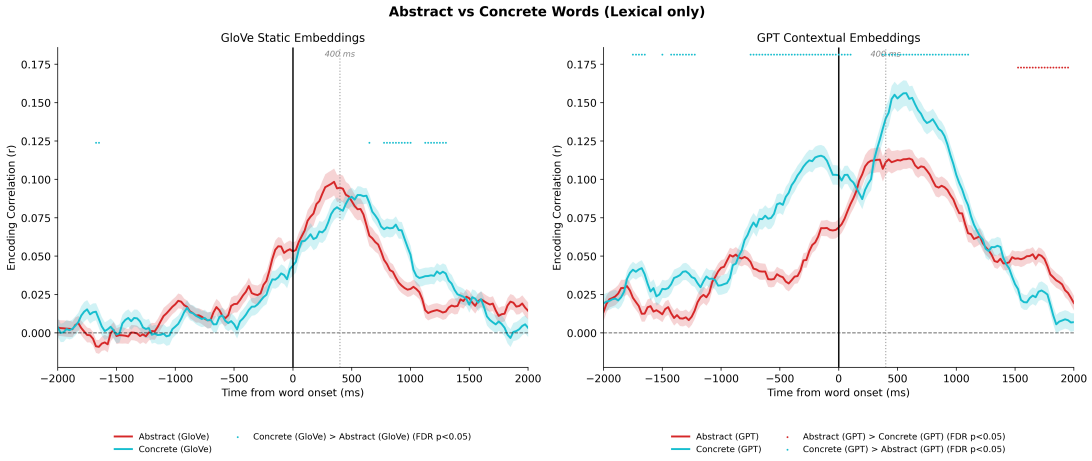
The baseline comparison (see appendix figure 11) confirms that contextual embeddings outperforms static embeddings with GPT's embedding having a higher correlation throughout. This higher correlation is sustained regardless of word category (figure 7), further validating the claim the brain represents words contextually.



(a) Verbs vs Nouns



(b) Function vs Content Words



(c) Abstract vs Concrete words

Figure 7: Subplot of the results from the second hypothesis: encoding performance based on word categorisation. Left-side shows Glove embedding, right side shows GPT embedding. Asterisks indicate time points where one category significantly outperformed the other in encoding correlation (permutation test with FDR correction and a majority electrode threshold).

Nouns have a higher correlation than verbs for both embeddings (figure 7 subplot a) with

GloVe peaking at 400ms whereas, GPT presents a later and more sustained peak likely due to it's contextual representation. The higher correlation can be explained by nouns having a more distinct and stable semantic representation and embedding space compared to verbs, allowing for stronger brain to model alignment.

The pre-onset divergence for nouns versus verbs, is not seen for function and content words (figure 7 subplot b). However, post-onset, content words have significantly higher correlation compare to function words regardless of embedding. This likely reflects greater processing demands to content words as they have richer semantic meaning which is consistent with current literature [9].

Unlike the negligible difference in GloVe, GPT embedding showed significantly higher correlation for concrete words compared to abstract words (See figure 7 subplot c). The difference can again be explained by GPT use of contextual embeddings enabling stronger representation and prediction of upcoming words, which may contribute to larger differences in post word onset. The higher correlation for concrete words may reflect greater neural activation for sensory grounded words compared to words that do not trigger strong sensory associations.

Therefore, the results support the second hypothesis, as the correlation between the neural signal and word embeddings varied across linguistic categories.

Conclusion

This study replicated key findings from the paper while explaining observed variations. Additionally, the study showed semantic distance and word category modulate encoding model correlations, complementing predictive coding theory.

References

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Appendix

Below are the additional figures used for the analysis. To see key brain area's that were actived for the second hypothesis: based on word categorisation, look at the Jupter notebook Novel Hypothesis.

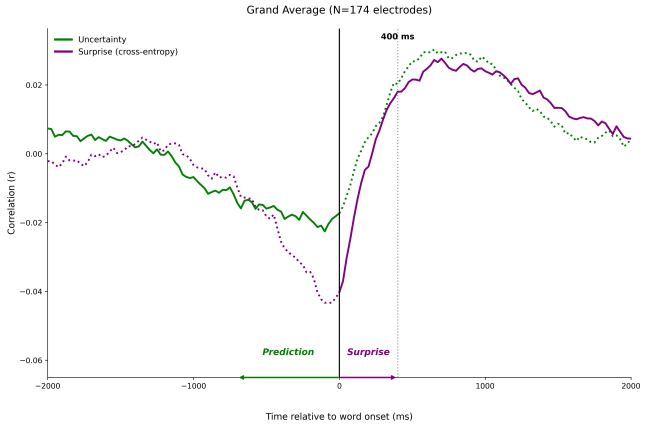


Figure 8: Replication for figure 5b from paper but using Pearson Correlation. All 174 electrodes were used.

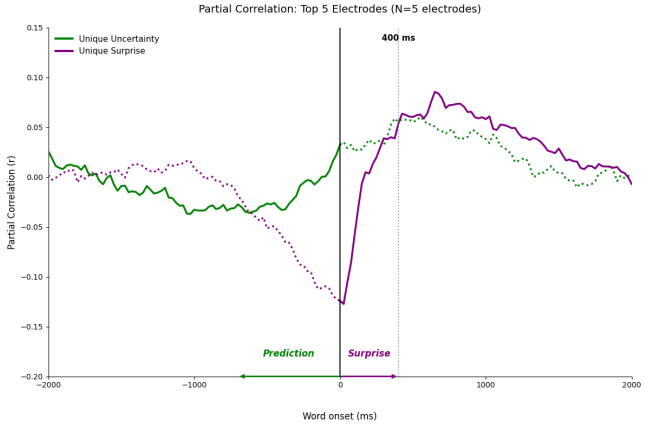


Figure 9: Replication for figure 5b from paper but only a subset of 5 electrodes. Correlations are partial correlations.

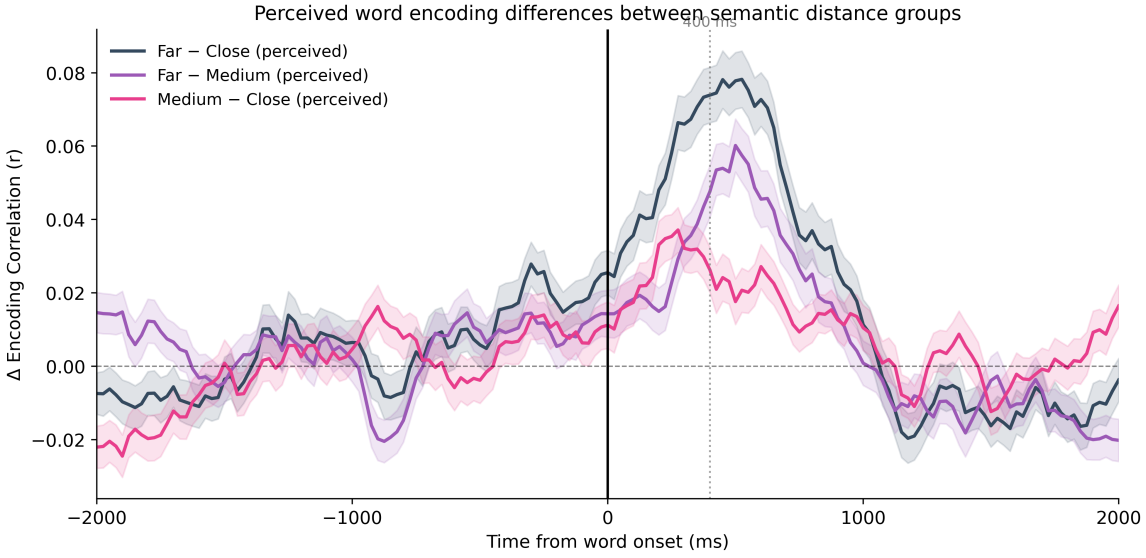


Figure 10: Three pairwise difference curves (FarClose, FarMedium, MediumClose) were plotted over time. Grand-averaged encoding correlations show stronger neural responses for semantically distant word pairs after onset, peaking around 400 ms. The largest effect (Far–Close) suggests semantic distance is primarily encoded post-onset.

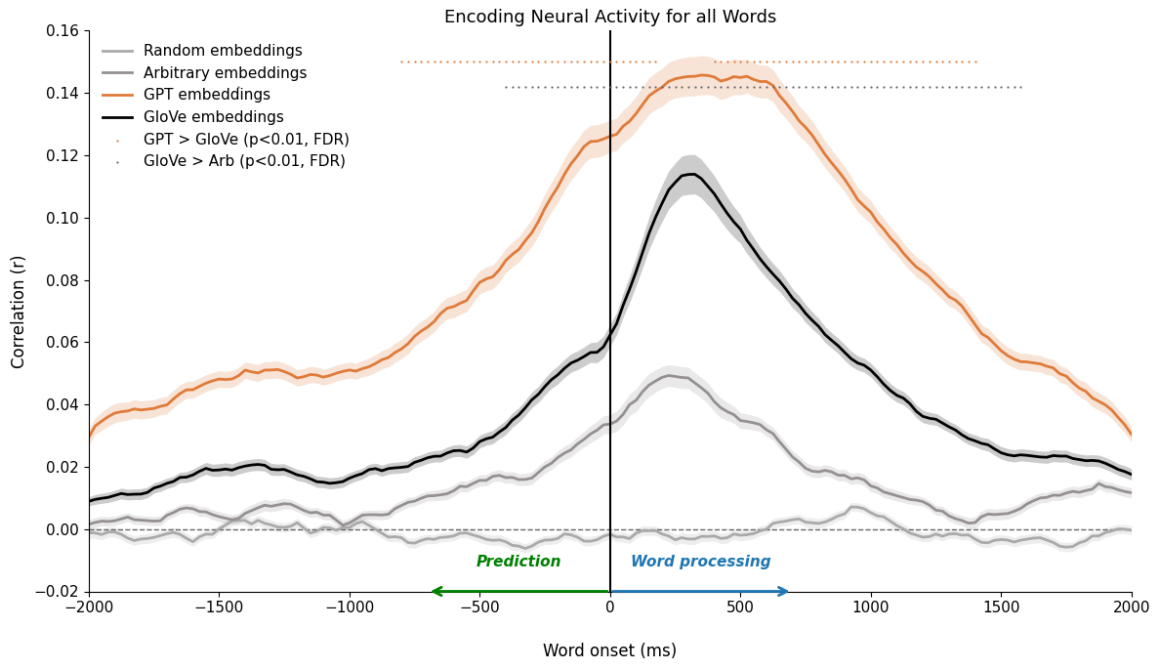


Figure 11: Replication of figure 4a from the paper with the inclusions of the GPT embedding. Figure demonstrates baseline; when using all words, the GPT embedding has the highest correlation.